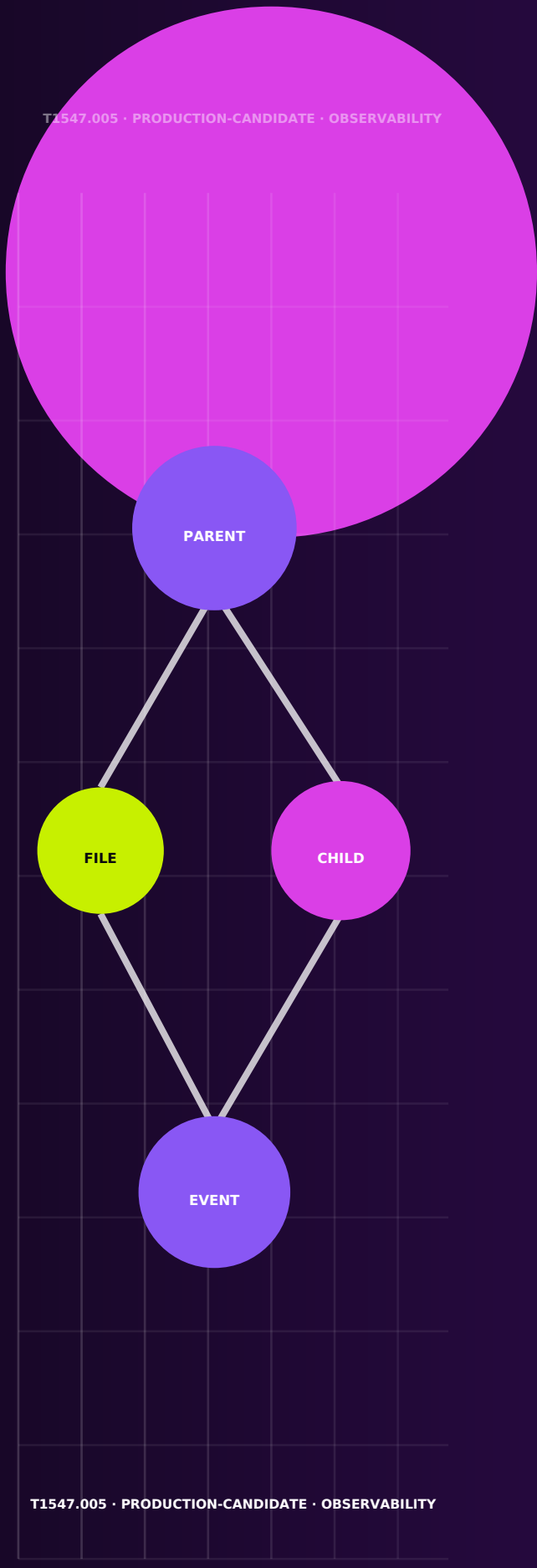
DETECTION THREAT LANDSCAPE

MEDUSA - MIMILIB LSA SECURITY PACKAGE

Source-observed registration of the Mimikatz mimilib Security Support Provider for plaintext credential capture.



PRODUCTION CANDIDATE

CONFIDENCE: MEDIUM

REPORT TYPE
campaign

STATUS
Untested implementation sketch

PRIMARY PLATFORM
Microsoft Defender XDR /

OPERATIONAL VALUE
9 / 10

01

Executive Assessment

Decision-grade summary with explicit evidence boundaries.

ASSESSMENT

DECISION

Retain this package as a production candidate and migrate it to the canonical visual and evidence contract without changing query conditions. Treat every result as an investigative lead until local schema, field population, benign overlap and the legacy claim provenance are reviewed.

LEGACY EVIDENCE RECORD

- The retained legacy source ledger contains 4 normalized references; claim-to-source mapping remains...

ANALYST ASSESSMENT

- The existing logic is retained at production-candidate maturity; repository migration is not...
- The folder and query header map the package to T1547.005 / T1003.001; operational relevance...

DETECTION ASSUMPTIONS

- The named telemetry is collected across the intended asset or identity scope with adequate...
- Projected fields preserve their expected semantics in the target connector and parser. [C002]
- Approved activity, owners and change records are available to analysts during disposition. [C003]

KEY FINDINGS

FINDING 1

The original query conditions and source ledger were preserved during layout migration; legacy PDF SHA-256 fcd94c9b67e7862e... is recorded for traceability. [C001]

FINDING 2

The principal observable is Evaluate DeviceRegistryEvents for mimilib lsa security package using the existing source-derived conditions while preserving projected investigation context. [C002]

FINDING 3

The evidence boundary remains explicit: a match does not prove Detection boundary: a match proves the Registry value contained "mimilib"; it does not prove the DLL loaded, passwords were captured, or Medusa attribution. [C003]

EVIDENCE BOUNDARY

A matching result does not prove Detection boundary: a match proves the Registry value contained "mimilib"; it does not prove the DLL loaded, passwords were captured, or Medusa attribution. Target-environment field population, benign prevalence, parser behavior and operational ownership have not been validated by this repository migration.

02

Threat Mechanics

Technical mechanism, observable transitions and supported scope.

ATTACK FLOW

TECHNICAL BACKGROUND

The retained dossier links Medusa to the detection opportunity described as mimilib lsa security package. The v2 migration preserves the query exactly and exposes its match boundary, but does not reinterpret legacy source claims as laboratory evidence. Evaluate DeviceRegistryEvents for mimilib lsa security package using the existing source-derived conditions while preserving projected investigation context. [C001][C002]

1

THREAT CONTEXT

Source-observed registration of the Mimikatz mimilib Security Support Provider for plaintext credential capture. Observable: Context retained from the legacy dossier and source ledger.

2

OBSERVABLE CANDIDATE

Evaluate DeviceRegistryEvents for mimilib lsa security package using the existing source-derived conditions while preserving projected investigation context. Observable: Events from DeviceRegistryEvents and projected fields.

3

CONTEXTUAL ENRICHMENT

Resolve asset, identity, process, ownership and change context before disposition. Observable: Raw event, surrounding timeline, inventory and authorized-change evidence.

4

DECISION GATE

Escalate only when corroborating evidence closes the documented non-claim boundary. Observable: Controlled validation, explainable negatives and analyst-readable evidence.

03

Telemetry & Evidence Model

Minimum data contract and known visibility gaps.

DATA CONTRACT

PRIMARY PLATFORM

Microsoft Defender XDR /. Evaluate DeviceRegistryEvents for mimilib lsa security package using the existing source-derived conditions while preserving projected investigation context.

FIELD / TABLE	STATUS	PURPOSE
DeviceRegistryEvents	**Required**	Primary event source selected by the unchanged query logic.
Event time and entity identifiers	**Required**	Chronology, deduplication, scoping and investigation pivots.
Raw projected context	**Required**	Preserve the fields emitted by the query before normalization or tuning.
Asset, identity and change inventory	**Enrichment**	Separate authorized activity and define operational ownership.

KNOWN VISIBILITY GAPS

Unvalidated local schema Missing projected fields Incomplete retention or asset scope Mutable attacker behavior Legacy claim provenance review pending

04

Primary Detection Logic

Native query, explicit condition evidence and match meaning.

KQL

```
// Rule: [T1547.005][T1003.001] Medusa - Mimilib LSA Security Package
// MITRE ATT&CK: T1547.005, T1003.001
// Classification: Production candidate
// Status: Untested implementation sketch
// Primary platform: Microsoft Defender XDR /
// Required telemetry: DeviceRegistryEvents
// Purpose: Source-observed registration of the Mimikatz mimilib Security Support Provider for plaintext credential capture.
// Logic: Evaluate DeviceRegistryEvents for mimilib lsa security package using the existing source-derived conditions while preserving projected investigation contex
// A match proves: Detection boundary: a match proves the Registry value contained "mimilib"; it does not prove the DLL loaded, passwords were captured, or Medusa at
// A match does not prove: Detection boundary: a match proves the Registry value contained "mimilib"; it does not prove the DLL loaded, passwords were captured, or M
// Expected false positives: authorized administration, approved software, testing, or benign artifact collisions.
// Coverage limits: local schema, field population, retention, collection scope, and attacker variation require validation.
// Evidence claims: C001, C002, C003
//
// Production candidate - Untested implementation sketch
// Primary platform: Microsoft Defender XDR / Advanced Hunting
// Required telemetry: Microsoft Defender for Endpoint registry events on Windows
// Evidence: CISA, FBI and HHS report Medusa actors adding mimilib to the LSA
// Security Packages REG_MULTI_SZ value so the credential-stealing SSP loads in LSASS.
// Detection boundary: a match proves the Registry value contained "mimilib"; it does
// not prove the DLL loaded, passwords were captured, or Medusa attribution.
DeviceRegistryEvents
| where RegistryKey endswith @"\SYSTEM\CurrentControlSet\Control\Lsa"
| where RegistryValueName =~ "Security Packages"
| where RegistryValueData contains "mimilib"
| project
    Timestamp,
    DeviceId,
    DeviceName,
    ReportId,
    ActionType,
    RegistryKey,
    RegistryValueName,
    RegistryValueType,
    RegistryValueData,
    PreviousRegistryValueData,
    InitiatingProcessFileName,
    InitiatingProcessFolderPath,
    InitiatingProcessCommandLine,
    InitiatingProcessAccountName,
    InitiatingProcessSHA1,
    InitiatingProcessUniqueId
| order by Timestamp desc
```

WHAT A MATCH PROVES

Detection boundary: a match proves the Registry value contained "mimilib"; it does not prove the DLL loaded, passwords were captured, or Medusa attribution.

WHAT IT DOES NOT PROVE

Detection boundary: a match proves the Registry value contained "mimilib"; it does not prove the DLL loaded, passwords were captured, or Medusa attribution.

MATURITY

Production candidate - Untested implementation sketch. Evaluate DeviceRegistryEvents for mimilib lsa security package using the existing source-derived conditions while preserving projected investigation context.

05

Triage & Hunting Playbook

Evidence-preserving workflow from match to disposition.

WORKFLOW

01

Preserve the complete raw matching event, query version, timestamp, connector or sensor version and stable entity identifiers.

02

Confirm the event originated from the intended Microsoft Defender XDR / data source and verify the semantics of every condition used by the query.

03

Resolve asset, identity, process, software and destination ownership against authoritative inventories and approved change records.

04

Build the surrounding endpoint, identity, network or appliance timeline and seek an independent evidence layer before escalation.

05

Classify benign overlap explicitly; tune only with durable ownership or provenance controls, never by suppressing the core behavior globally.

06

Escalate only when corroborating evidence establishes malicious or unauthorized activity beyond the documented match boundary.

DISPOSITION RULE

Escalate only when the evidence establishes the behavior described by the query. Detection boundary: a match proves the Registry value contained "mimilib"; it does not prove the DLL loaded, passwords were captured, or Medusa attribution.

06

False Positives & Validation

Overlap model, controlled checks and promotion gate.

QUALITY GATE

AUTHORIZED ADMINISTRATION OR SOFTWARE

Require owner, purpose, approved change and expected execution or configuration provenance.

BENIGN ARTIFACT OR COMMAND COLLISION

Require the full condition set plus an independent evidence layer or stable provenance.

PARSER, INVENTORY OR RETENTION AMBIGUITY

Reproduce a controlled event and reconcile raw fields, collection scope and authoritative inventory.

VALIDATION SEQUENCE

- Generate an authorized controlled equivalent of the primary observable without using live malicious infrastructure.
- Verify each query condition and projected field against retained raw telemetry.
- Run representative approved and near-match negative controls.
- Measure historical prevalence and classify every returned owner or workflow.
- Record parser, connector, retention and asset-scope dependencies with the test evidence.

SUCCESS

The controlled positive produces a complete, explainable row and representative negatives are separable through durable context.

REJECTION

Reject promotion when field semantics, collection scope, benign ownership or the core behavior cannot be established reproducibly.

PROMOTION GATE

Require claim-level source review, a retained controlled positive, representative negatives, measured prevalence and an accountable operational owner.

07

Analyst Assessment & References

Coverage statement, open questions and source record.

FINAL

CONCLUSION

This package now uses the canonical v2 layout and evidence contract while preserving the original detection.kql conditions. It remains a production candidate at medium confidence. The visual migration is complete; claim-level provenance and target-environment execution remain explicit promotion gates.

MITRE ATT&CK

T1547.005 / T1003.001

PRIMARY AND AUTHORITATIVE REFERENCES

S001 ic3.gov - Updated with FBI investigations through 2026-04. (2025-03-12)

hxxps://www[.]ic3[.]gov/CSA/2026/260818[.]pdf

S002 Microsoft Learn - Microsoft Learn, "DeviceRegistryEvents table in the advanced hunting schema. (2025-06-11)

hxxps://learn[.]microsoft[.]com/en-us/defender-xdr/advanced-hunting-deviceregistryevents-table

S003 MITRE ATT&CK - MITRE ATT&CK, T1547.005, "Boot or Logon Autostart Execution: Security Support Provider. (2025-06-11)

hxxps://attack[.]mitre[.]org/techniques/T1547/005/

S004 MITRE ATT&CK - MITRE ATT&CK, T1547.005, "Boot or Logon Autostart Execution: Security Support Provider. (2025-06-11)

hxxps://attack[.]mitre[.]org/techniques/T1003/001/

OPEN QUESTIONS

A matching result does not prove Detection boundary: a match proves the Registry value contained "mimilib"; it does not prove the DLL loaded, passwords were captured, or Medusa attribution. Target-environment field population, benign prevalence, parser behavior and operational ownership have not been validated by this repository migration.

DOCUMENT CONTROL

Schema 2.0 | Layout 2.0 | Maturity production-candidate | Confidence Medium | No customer data | Evidence ledger required